

AI Engineer Roadmap for Rakesh

Completed

✓ Python

- Variables
- Data Types
- Strings
- Lists
- Dictionaries
- Functions
- OOP
- File Handling
- JSON
- Regex
- Try Except

✓ NumPy

- Arrays
- 2D Arrays
- Statistics
- Matrix Operations

✓ Pandas

- Series
- DataFrame
- CSV
- Filtering
- GroupBy
- Merge

✓ Matplotlib

- Line Chart
- Bar Chart
- Scatter Plot
- Histogram
- Pie Chart

✓ OpenAI API

- Chat API
- Structured Output

- Function Calling
 - Streaming
-

PART 1 - AI / Machine Learning

Phase 3 - Scikit-learn

Learn:

- Train/Test Split
- Linear Regression
- Multiple Regression
- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- K-Means
- Confusion Matrix
- Cross Validation
- Grid Search

Purpose:

- House Price Prediction
- Spam Detection
- Loan Approval
- Customer Churn

Study:

<https://scikit-learn.org/stable/>

Easy Course:

<https://www.kaggle.com/learn/intro-to-machine-learning>

Phase 4 - PyTorch

Learn:

- Tensor
- Dataset
- DataLoader

- Neural Networks
- Training Loop
- Optimizer
- Loss Function

Purpose:

- Deep Learning
- Image Classification
- Neural Networks
- Transformers

Study:

https://pytorch.org/tutorials/beginner/deep_learning_60min_blitz.html

Phase 5 - Transformers

Learn:

- Attention
- Self Attention
- Encoder
- Decoder
- BERT
- GPT
- Llama
- Tokenizers

Purpose:

Technology behind:

- ChatGPT
- Claude
- Gemini
- Llama

Study:

<https://huggingface.co/learn/llm-course/chapter1/1>

PART 2 - Generative AI

Phase 6 - OpenAI API

✓ Completed

Learn:

- Chat API
- Structured Output
- Function Calling
- Streaming

Purpose:

- AI Chatbot
- AI Code Reviewer
- Resume Analyzer
- AI Assistant

Docs:

<https://platform.openai.com/docs/overview>

Phase 7 - Embeddings

Learn:

- Text to Vector
- Semantic Search
- Similarity Search
- Vector Search

Purpose:

- Semantic Search
- Similar Documents
- RAG

Docs:

<https://platform.openai.com/docs/guides/embeddings>

Phase 8 - RAG

Learn:

- PDF Loading
- Chunking
- Embeddings
- Vector Database
- Retrieval
- Question Answering

Purpose:

- AI PDF Chat
- Company Knowledge Base
- Legal Documents
- Medical Documents

Flow:

PDF

↓

Chunking

↓

Embeddings

↓

Vector Database

↓

GPT

↓

Answer

Study:

<https://python.langchain.com/docs/tutorials/rag/>

Phase 9 - LangChain

Learn:

- PromptTemplate
- Chains
- Tools
- Memory
- RAG

Purpose:

Connect:

- GPT
- Embeddings
- Vector DB
- Tools
- Memory

Study:

<https://python.langchain.com/docs/introduction/>

Phase 10 - AI Agents

Learn:

- Tool Calling
- Memory
- Planning
- Reasoning
- Multi-step Execution
- Multi-Agent Systems

Purpose:

Build:

- AI Assistant
- Job Search Agent
- Email Agent
- Research Agent

Study:

Most Important Order

Python ✓

NumPy ✓

Pandas ✓

Matplotlib ✓

OpenAI API ✓

↓

Embeddings

↓

RAG

↓

LangChain

↓

AI Agents

↓

Scikit-learn

↓

PyTorch

↓

Transformers

Best Projects

1. AI Chatbot
2. AI PDF Chat (RAG)
3. AI Code Reviewer ★★★★★
4. AI Resume Analyzer
5. AI Agent with Tools

Goal:

Become an AI Engineer / Generative AI Engineer with strong software engineering and backend expertise.